Instance-Aware Spatial-Frequency Feature Fusion Detector for Oriented Object Detection in Remote-Sensing Images

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***Abstract*— In recent years, fusing multitype features poses great potential for oriented object detection (OOD) in remote-sensing images (RSIs). Due to the inexplicit operation of modeling orientation variations, convolutional neural networks (CNNs) are difficult to perceive objects under different transformations (angles and scales). In this article, we propose a novel instance-aware spatial-frequency feature fusion detector (SFFD) for OOD in RSIs. First, a layerwise frequency-domain analysis (L-FDA) module is built along with CNN layers to extract frequency features. Getting rid of the constraints such as horizontal rectangular kernels in CNNs, our L-FDA possesses an outstanding ability to locate mutational signals from frequency space. These mutational signals record the scale and angle information of the oriented instances in images. Subsequently, CNN and frequency features are sent into the region-of-interest (RoI) Pooling layer to obtain multitype instance-level RoI features. Moreover, the proposed instance-aware cross-feature fusion (CFF) module explores the interaction between these diverse features which provides an explicit indicator to compensate for the orientation information ignored by instance-level CNN features. Finally, our SFFD unifies the proposed L-FDA module and CFF module into the detection network to localize oriented instances in RSIs. We compare our method with many state-of-the-art methods on DOTA, HRSC2016, and NWPU VHR-10 datasets. Experimental results verify the validity of modeling instance-level object relations from the frequency domain and CNNs for OOD.**

***Index Terms*— Convolutional neural networks (CNNs), feature fusion, frequency-domain analysis, oriented object detection (OOD), remote-sensing images (RSIs), spatial-frequency feature.**

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I. Introduction

With the recent advances in sensors and Earth observation techniques, image processing of remote-sensing images (RSIs) [1], [2], [3], [4], [5], [6], [7] have attracted much attention in recent years, including oriented object detection (OOD). Different from object detection in natural images, OOD in RSIs is more challenging due to the dense location, various scales, arbitrary orientations, and large aspect ratio of target objects. The rapid development of convolutional neural networks (CNNs) has greatly accelerated the progress of many computer vision tasks such as image classification [8], [9], [10], [11], change detection [12], [13], and OOD [14], [15], [16].

With the limitation of a horizontal rectangular filter, ordinary CNNs face great difficulties in perceiving objects under different angles and scales. Most existing OOD methods focus on generating oriented proposals or anchors [17], [18], [19] to better represent oriented objects and rotation angles of instances are obtained by the user-defined or learnable parameters. However, these methods cause a massive memory footprint and excessive computation costs. Many recent methods have been proposed to excavate better features to improve the performance of OOD in RSIs [20], [21], [22], [23]. They utilized an attention mechanism and instance-aware relation learning strategy to aggregate the context information to highlight the key spatial features and suppress the redundant ones. Recent researches focus on extracting the rotation-invariant features [24], [25], [26], [27] to improve the performance of CNNs in modeling rotation variations. They design an explicit regularization constraint to align the spatial and channel rotation-invariant features. Although the oriented proposals or anchors generation strategy is a valid choice to eliminate the inconsistency between the localization and classification tasks caused by the incomplete receptive fields. Abundant proposals and anchors bring massive memory footprint and computation costs. Using an attention mechanism to extract high-quality features significantly improves the performance of small object detection, but cannot obtain the rotation angle of objects. Moreover, modeling the rotation-invariant features also needs more parameters to learn the rotation-invariant transformation which results in a significant increase in computation.

Our approach is motivated by the success of multitype feature fusion in many computer vision tasks [28], [29], [30],

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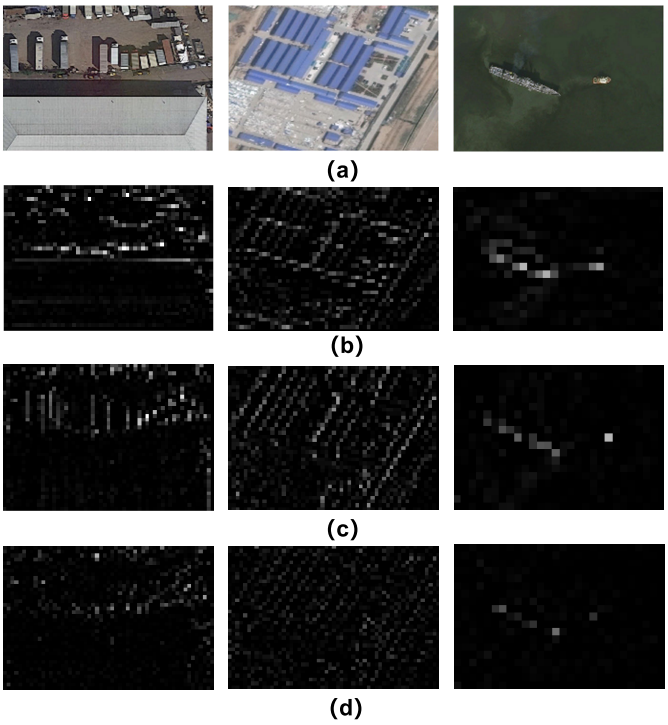


Fig. 1. Visualization of frequency features extracted by the Haar WPT. (a) Examples of original images. (b) Corresponding horizontal features. (c) Corresponding vertical features. (d) Corresponding diagonal features.

[31]. Exploring different properties to describe the same object and excavating the interaction between these properties poses great potential for different tasks. Most feature fusion strategies are inclined to fuse features in image-level [29], [30], [31] which is suboptimal to integrate low-level features (e.g., edge information) extracted from the frequency-domain analysis with high-level semantic CNN features. A deeper feature extraction network in OOD can further exacerbate this misalignment. To alleviate the limitation above, frequency domain orientation learning (FDOL) [30] proposed an orientation-enhanced self-attention layer (OES-Layer) to integrate frequency features with image-level CNN features. However, they unilaterally considered the influence of frequency information on spatial features which ignored the interaction relationship between the two sides. Thus, our proposed SFFD aims for excavating the comprehensive interaction among multitype instance-level features to improve the performance of OOD. As illustrated in Fig. 1, we visualize the features extracted by the Haar wavelet packet transform (WPT). We can clearly see that the edge information of instances with different transformations (scales and angles) are all captured.

In this article, we try to explore an intuitive connection among instance-level CNN features, frequency features, and the combined spatial-frequency features as an additional joint constraint for training the detector. First, the proposed instance-aware spatial-frequency feature fusion detector (SFFD) introduces a layerwise frequency-domain analysis (L-FDA) module, which is built along with the CNN layers to extract frequency features, synchronously. The proposed L-FDA iteratively extracts multiscale frequency features that are consistent with the backbone feature scale. Then, the CNN features and frequency features are fed into different region-of-interest (RoI) Pooling layers to obtain multitype fixed-size RoI features. Moreover, we propose a cross-feature fusion (CFF) module to explore the interaction between these diverse features which provides an explicit indicator to compensate for the orientation information ignored by CNNs. Finally, we unify the L-FDA and CFF into a complete detector (SFFD) to detect oriented instances in RSIs. In summary, the innovative contributions of this article are as follows.

1) We introduce an instance-aware CFF module to extract orientation information for perceiving rotated instances from CNN and frequency domains. Exploring the interaction between these diverse features gives CNNs a powerful potential to perceive different rotated instances in RSIs.
2) The proposed L-FDA unifies WPT along with the convolutional layer to extract CNN and frequency features, synchronously. Our L-FDA locates the mutational signals in the frequency space. These signals provide an explicit indicator to compensate for the orientation information ignored by CNN features.
3) We unify the L-FDA module with the backbone extractor and the proposed CFF is integrated into the detector network. Compared with other state-of-the-art methods, our instance-aware SFFD achieves competitive performance on DOTA, HRSC2016, and NWPU VHR-10 datasets.

The rest of this article is organized as follows. We review recent works related to our method in Section II and the details of our method are introduced in Section III. Section IV summarizes the results achieved by our model and the comparisons with many state-of-the-art approaches are also listed. Finally, the conclusion of this article and many remarks for the further works are given in Section V.

II. Related Work

A. Oriented Object Detection in RSIs

In RSIs, OOD has received extensive attention due to its wide applications. Commonly, there are two definitions of the oriented bounding box (OBB): the five-parameter regression method and the eight-parameter regression method. However, these definitions suffer from boundary discontinuity problems caused by improper angular periodicity and corner ordering. Thus, Yang and Yan [32] proposed a circular smooth label (CSL) technique that transformed the angle prediction problem from a regression task to a classification one. The densely coded label (DCL) module was introduced to solve the excessive computation cost caused by CSL. In order to eliminate the mismatches between the RoIs and oriented objects, the RoI transformer [17] transferred horizontal RoIs to the rotated ones so that oriented instances can be covered by the rotated bounding boxes. Xie et al. [19] proposed a region proposal network (O-RPN) to generate high-quality oriented anchors to better perceive the target instances. The midpoint offset representation strategy was introduced to reduce the number of parameters and decrease the risk of overfitting.

Many methods tend to highlight the most discriminative regions in images. Zhang et al. [20] proposed multiscale

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attention-enhanced features to capture the global and local contextual spatial information to locate the target instances. The spatial correlations among oriented objects are effectively collected at scene-level and object-level features. Wang et al. [21] constructed an area-weighted loss function to capture the response of small objects. The context information, in both multiple scales and same scale feature maps, was captured by the proposed feature-merged single-shot detection (FMSSD) network. Ming et al. [22] proposed a polarization attention module (PAM) to decouple the classification and regression features. The constructed robust critical features are adaptively refined to improve the performance of oriented object localization. Fu et al. [23] designed a top-down pathway to augment the features of the shallow layer with semantic representations. The proposed feature-fusion architecture combined different levels of features to obtain a powerful representation for multiscale instances.

Many types of research focus on modeling the interaction among multitype features to improve the performance of OOD in RSIs. Wu et al. [28] for the first time proposed a rotation-invariant spatial-frequency detector for OOD. Under the supervision of the theoretical proof of the rotation invariance, the performance of OOD is significantly improved. Zheng et al. [30] proposed a variant of self-attention architecture that utilized the proposed frequency-domain feature extraction network (FFE) to extract orientation features. An orientation weight was constructed by the orientation features to refine the interactions among spatial locations. However, this variant of self-attention architecture was integrated into the backbone network multiple times which brings more computational overhead. Moreover, Zheng et al. [30] unilaterally considered the influence of frequency information on spatial features which ignored the interaction relationship between the two sides.

B. Multitype Feature Extraction

Using different information or properties (e.g., contexts, resolution, sensors, and so on) to describe the same object and exploring the interaction between these properties is an efficient way to improve the performance of many computer vision tasks [29], [30], [31], [33]. Thus, much research is devoted to constructing different multitype feature extraction modules. In order to excavate different discriminative features from raw HSIs, Roy et al. [29] utilized two types of convolution layers to construct multitype features. These features were integrated via concatenation operation to boost the representation of backbone features. Zhang and Zhang [33] proposed weighted distance tools to extract hand-crafted features and these traditional features are injected into CNNs to classify the ships in SAR images. SpectralNET [31] constructed a Spatial-Spectral WaveletCNN to extract low-level frequency edge features. These features were concatenated with backbone features to bring out multiscale feature vectors. Prabhakar and Geetha [34] proposed a 2-D empirical wavelet transform (EWT) to alleviate the class imbalance problem caused by the limited availability of training samples.

C. Multitype Feature Fusion

Fusing multitype features is a valid fashion to alleviate the weak feature representation caused by the single context information [8], [28], [30], [32], [35], [36], [37]. Hong et al. [8] proposed a general multimodal deep-learning (MDL) framework to address the poor performance of single-modality-domain classification networks caused by the limitation of information diversity. They explored different forms to fuse multitype features and unified many plug-and-plug fusion modules into their MDL network. Tu et al. [36] constructed bidirectional features transformation-based joint learning framework to optimize the boundary and region features simultaneously. The multiscale boundary features were embedded into instance-level features to propagate high-quality semantic information. Liu et al. [38] utilized CNNs to extract visual information and the multitype features to obtain multimodal samples. Two kinds of heterogeneous features are concatenated in the channel dimension to measure the related information among multimodal samples. However, most of the above-mentioned methods [28], [30], [31], [34] explore the interaction relation among multimodal features from the data level or image level. It is a nontrivial task to use these multimodal feature fusion fashions for OOD in RSIs.

III. Proposed Method

The proposed instance-aware SFFD starts with an L-FDA module. Our L-FDA is built along with the feature extractor network, aiming for extracting the frequency features. Different from the image-level features extracted by CNNs, frequency features provide an explicit indicator to compensate for the orientation information ignored by CNNs. After feeding spatial-frequency features into different RoI Pooling layers, the proposed CFF module comprehensively considers the interactions between these instance-level RoI features. Object relations over these multitype instance-level features are aggregated and propagated to refine the CNN RoI features. Finally, the enhanced RoI features are fed into the detector to obtain more precise detection results of OOD in RSIs. The overall of our SFFD is shown in Fig. 2.

A. Layerwise Frequency-Domain Analysis

We utilize multilevel WPT as a base tool in the proposed L-FDA to decompose different frequency components for the following reasons. First, WPT is a widely used technique in signal processing, including digital images. From the frequency space, WPT enables to locate mutational signals which record the scale and angle information of target-oriented instances. Second, frequency features extracted by WPT contain rich patterns for image understanding tasks. These patterns can be considered as an explicit indicator to compensate for the orientation information ignored by the CNN features

$$F_{\mathrm{fd}} = \mathbf{WPT}(\mathcal{X}) \tag{1}$$

$$\mathcal{X} = \mathbf{WPT}_{\mathrm{inverse}}(F_{\mathrm{fd}}) \tag{2}$$

where F_{fd} and $\mathcal{X}$ denote the frequency-domain features and original image. WPT represents the wavelet package transform and $\mathrm{WPT}_{\mathrm{inverse}}$ denotes the inverse transform.

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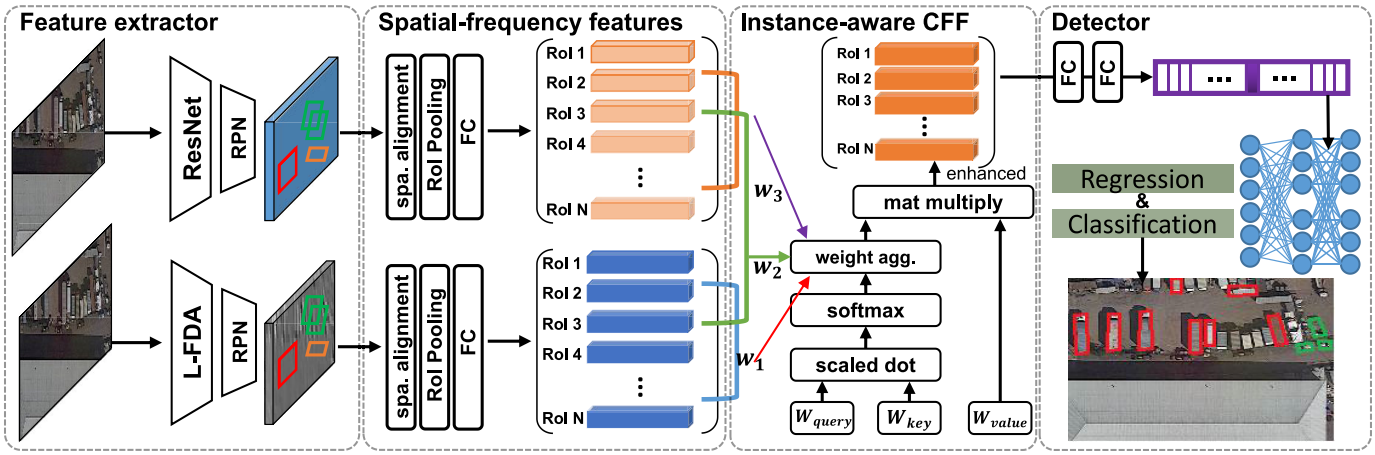


Fig. 2. Pipeline of our instance-aware SFFD. First, the proposed L-FDA module is built along with the CNN backbone extractor to obtain frequency features. The CNN and frequency features are fed into the spatial alignment and RoI Pooling layer to obtain multitype instance-level RoI features. Subsequently, the proposed CFF module explores the interaction between these diverse features. Finally, the enhanced instance-level CNN RoI features are fed into the detector to make the category classification and object localization.

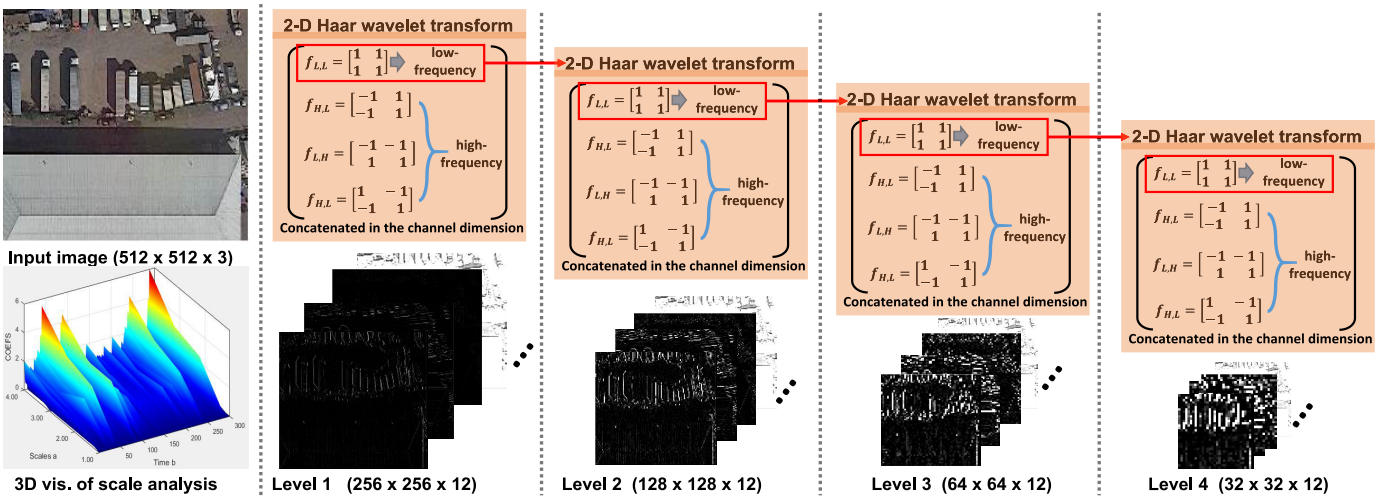


Fig. 3. Visualization of our L-FDA. Multilevel WPT is adopted in L-FDA to extract multiscale frequency features. The 3-D visualization of the scale analysis proves the necessity of the multiscale strategy in L-FDA. The extracted frequency features are fed into the following RoI Pooling layer.

The multilevel WPT used in our method contains four filters: $f_{L,L}$, $f_{L,H}$, $f_{H,L}$, and $f_{H,H}$. Three high-frequency information (horizontal, vertical, and diagonal edge features) and one low-frequency information are concatenated in the channel dimension. Considering that features, extracted from the CNN extractor, have four different scales. The proposed L-FDA constructs a four-level WPT to achieve the multiscale frequency features. Conceptually, we set the scale factor to 2 and a total of four-level of WPTs is constructed in our L-FDA. The definition of $f_{L,H}$, $f_{H,L}$, and $f_{H,H}$ are as follows:

$$f_{L,H}=\begin{bmatrix}-1 & -1\\ 1 & 1\end{bmatrix},\quad f_{H,L}=\begin{bmatrix}-1 & 1\\ -1 & 1\end{bmatrix},\quad f_{H,H}=\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}. \tag{3}$$

The frequency features can be obtained by

$$F_{h,s}, F_{v,s}, F_{d,s}, F_{ds,s}=\begin{cases}\mathbf{WPT}(\mathcal{X}, s), & s=1\\ \mathbf{WPT}\left(F_{ds,s-1}, s\right), & s=2,3,4.\end{cases} \tag{4}$$

The final frequency features F_{fd} is formulated as

$$F_{\mathrm{fd},s}=\left[F_{h,s}, F_{v,s}, F_{d,s}, F_{ds,s}\right],\quad s=1,2,3,4 \tag{5}$$

where [ ] denotes the concatenation operation.

As illustrated in Fig. 3, given an input image, we visualize different levels of frequency features and a 3-D view of scale analysis. We can clearly see that most of the mutation signals can be located at scale space level-2, level-3, and level-4. Considering that oriented objects always have different scales. The proposed L-FDA aims for extracting multiscale frequency features. Taking image $\mathcal{X}\in\mathbb{R}^{512\times512\times3}$ as input, the multilevel WPT extracts four scales of frequency features: $F_{\mathrm{fd},1}\in\mathbb{R}^{256\times256\times12}$, $F_{\mathrm{fd},2}\in\mathbb{R}^{128\times128\times12}$, $F_{\mathrm{fd},3}\in\mathbb{R}^{64\times64\times12}$, and $F_{\mathrm{fd},4}\in\mathbb{R}^{32\times32\times12}$. Finally, image-level frequency features F_{fd} and CNN features F_c are all fed into the CFF module.

B. Cross-Feature Fusion

Taking image-level frequency features F_{fd} and CNN features F_c into RoI Pooling layer P_1 and P_2, the coordinates of

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latent proposals are then used to obtain the instance-level RoI features. Note that the channel dimension of F_{fd} is different from F_c, we further utilize one fully connection layer FC to unify the channel dimension of these features as

$$r_f = \mathbf{FC}(\boldsymbol{P}_1(F_{\text{fd}})) \tag{6}$$

$$r_c = \boldsymbol{P}_2(F_c) \tag{7}$$

where r_c and $r_f \in \mathbb{R}^{N\times D}$. N and D denote the number and dimension of instance-level RoI features. Taking CNN RoI features $r_c \in \mathbb{R}^{N\times D}$ as an example, the enhanced features r_c^e can be formulated as

$$w_c = \text{softmax}\left(\frac{\boldsymbol{W}_{\text{query}}(r_c)\cdot \boldsymbol{W}_{\text{key}}(r_c)^T}{\sqrt{d_K}}\right) \tag{8}$$

$$r_c^e = w_c \cdot \boldsymbol{W}_{\text{value}}(r_c) \tag{9}$$

where W_{query}, W_{key}, and W_{value} denote three different *Linear* layers and $\cdot$ represents matrix multiplication. $w_c \in \mathbb{R}^{N\times N}$ records the relation among r_c which is the similarity weight needed for our CFF module.

Followed by method [39], geometry relation is introduced to model the positional affinity of two different regions which is formulated as

$$F_{\text{pos}} = \left\{\log\left(\frac{|x_a - x_b|}{w_a}\right),\ \log\left(\frac{|y_a - y_b|}{h_a}\right), \log\left(\frac{w_b}{w_a}\right), \times \log\left(\frac{h_b}{h_a}\right)\right\} \tag{10}$$

where x, y, w, and h denote center point coordinates, width, and height of instance regions r_c^a and r_c^b. After exploring all the positional relation among r_c, $F_{\text{pos}} \in \mathbb{R}^{N\times N\times 4}$ is embedded by

$$F_g = \begin{cases} \sin\left(F_{\text{pos}}(i)/1000^{2i/d}\right), & i = 0, 2 \\ \cos\left(F_{\text{pos}}(i)/1000^{2i/d}\right), & i = 1, 3 \end{cases} \tag{11}$$

where d represents the dimension of embedded features. Equation (11) denotes that the even-dimensional data in F_{pos} is embedded by *sine* function and the odd-dimensional data is embedded by *cosine* function. The geometry features $F_g \in \mathbb{R}^{N\times N\times d}$ is transferred by a set of CNN layers C and one reshape operation R to attain the geometry weight w_g

$$w_g = \boldsymbol{R}\left(\boldsymbol{C}\left(F_g\right)\right) \tag{12}$$

where $w_g \in \mathbb{R}^{N\times N}$. We further integrate geometry weight w_g with w_c

$$w_{gc} = \frac{w_g \cdot \exp(w_c)}{\sum_k w_{g,k}\cdot \exp\left(w_{c,k}\right)}. \tag{13}$$

Finally, (9) can be further formulated as

$$r_c^e = w_{gc} \cdot \boldsymbol{W}_{\text{value}}(r_c). \tag{14}$$

However, CNN RoI features do not distinctly model the orientation variations. This is determined by the nature of CNNs itself. Thus, the original similarity weight lacks the ability to perceive rotated instances. In order to alleviate the above-mentioned limitation, more diverse features are used to model the interaction relation among different oriented instances.

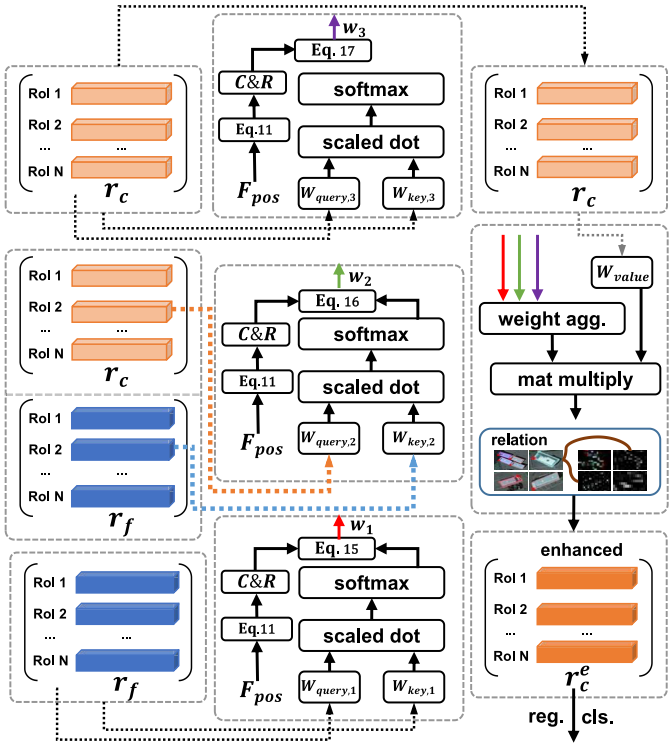


Fig. 4. Framework of our instance-aware CFF module. w_1, w_2, and w_3 denote three different similarity weights. *weight agg.* represents the operation of weight aggregation. *C&R* denotes a set of CNN layers and one reshape operation. F_{pos} is formulated in (10).

Fig. 4 illustrates the architecture of the proposed instance-aware CFF module which explores the interaction relationship among frequency r_f and CNN r_c RoI features to achieve a comprehensive similarity weight. The inputs are (r_c, r_f) and the outputs of CFF are enhanced CNN RoI features r_c^e

$$w_{\text{ff}} = \text{softmax}\left(\frac{\boldsymbol{W}_{\text{query},1}\left(r_f\right)\cdot \boldsymbol{W}_{\text{key},1}\left(r_f\right)^T}{\sqrt{d_{K_1}}}\right) \tag{15}$$

$$w_{\text{cf}} = \text{softmax}\left(\frac{\boldsymbol{W}_{\text{query},2}(r_c)\cdot \boldsymbol{W}_{\text{key},2}\left(r_f\right)^T}{\sqrt{d_{K_2}}}\right) \tag{16}$$

$$w_{\text{cc}} = \text{softmax}\left(\frac{\boldsymbol{W}_{\text{query},3}(r_c)\cdot \boldsymbol{W}_{\text{key},3}(r_c)^T}{\sqrt{d_{K_3}}}\right) \tag{17}$$

where w_{ff}, w_{cf}, and $w_{\text{cc}} \in \mathbb{R}^{N\times N}$ denote the relation weight calculated by the frequency, CNN–frequency, and CNN RoI features. $W_{\text{query},i}|_{i=1}^3$, $W_{\text{key},i}|_{i=1}^3$ denote six different *Linear* layers and $\cdot$ represents matrix multiplication.

Intuitively, r_f and r_c are attained by the same bounding box coordinates. Thus, we integrate w_{ff}, w_{cf}, and w_{cc} with the same geometry weight w_g [see (12)] as

$$w_1 = \frac{w_g \cdot \exp(w_{\text{ff}})}{\sum_k w_{g,k}\cdot \exp\left(w_{\text{ff},k}\right)} \tag{18}$$

$$w_2 = \frac{w_g \cdot \exp(w_{\text{cf}})}{\sum_k w_{g,k}\cdot \exp\left(w_{\text{cf},k}\right)} \tag{19}$$

$$w_3 = \frac{w_g \cdot \exp(w_{\text{cc}})}{\sum_k w_{g,k}\cdot \exp\left(w_{\text{cc},k}\right)}. \tag{20}$$

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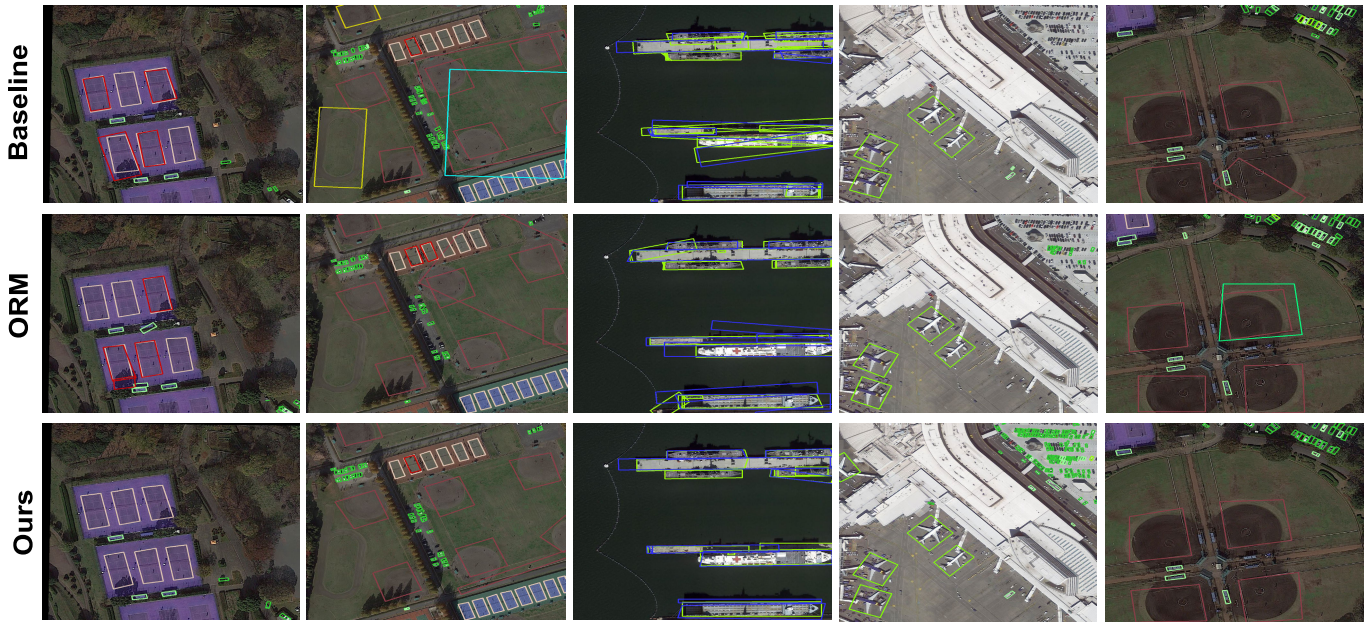


Fig. 5. Detection visualizations of baseline, ORM enhanced method, and our SFFD on DOTA. Compared with the base method, more miss-detected small vehicles and ships are detected by ORM enhanced method. Our SFFD further detects more targets and alleviates duplicate detection with different categories.

TABLE I
RESULTS OF DIFFERENT INSTANCE-LEVEL FEATURE FUSION METHODS ON DOTA TESTING SET

Methods	Backbone	Enhancer	mAP (%)	FPS	Parameter (M)
OPLD	R50	—	73.98	5.7	48.75
		ORM [39]	74.09	4.8	78.66
		CFF (Ours)	**74.21**	4.7	84.96
OPLD	R101	—	74.18	2.1	60.13
		ORM [39]	74.25	1.2	90.03
		CFF (Ours)	**74.67**	1.2	96.33
ReDet	ReR50	—	76.25	6.7	31.65
		ORM [39]	77.24	5.0	61.64
		CFF (Ours)	**78.01**	4.9	67.95

The final comprehensive similarity weight w is formulated as

$$w = \alpha_1 w_1 + \alpha_2 w_2 + \alpha_3 w_3 \tag{21}$$

where α_1, α_2, and α_3 denote the weighting parameter that are all set to 1.0 in our experiments. The enhanced CNN RoI features r_c^e in (9) can be further formulated as

$$r_c^e = w \cdot \boldsymbol{W}_{\text{value}}(r_c). \tag{22}$$

Finally, r_c^e is sent to the detector network for category classification and bounding box regression.

Integrated with detector: The proposed instance-aware CFF is integrated between the RoI Pooling layer and a common detector. Our CFF has the same input and output dimension which can be used in place within popular detection architecture (e.g., Faster RCNN [40], YOLOV2 [41], SSD [42], and so on) at an end to end training fashion.

IV. RESULT AND DISCUSSION

We evaluate the proposed SFFD on three public remote-sensing datasets: DOTA, HRSC2016, and NWPU VHR-10. The former two datasets are annotated by the OBBs and the last one is labeled by horizontal bounding boxes. Most compared methods are published in recent years.

A. Datasets and Implementation Details

1) DOTA: There are 2806 aerial images in DOTA dataset. The scale of images are range from 800 × 800 to 4000 × 4000. Fifteen categories of 188 282 samples are annotated in the DOTA dataset. The categories annotated in DOTA are as follows: Plane (PL), Baseball diamond (BD), Bridge (BR), Ground track field (GTF), Small vehicle (SV), Large vehicle (LV), Ship (SH), Tennis court (TC), Basketball court (BC), Storage tank (ST), Soccer-ball field (SBF), Roundabout (RA), Harbor (HA), Swimming pool (SP), and Helicopter (HC). Following the popular settings in previous methods [23], [24], [30], both training and validation sets are used for training. The rest images are used as a testing set.

We set the batch size to 2 per GPU and the stochastic gradient descent (SGD) optimizer with 0.9 momentum and 0.0001 weight decay is adopted in our experiments. The learning rate is set to 0.005 per GPU. Original images are cropped into 1024 × 1024 patches with an overlap of 200 pixels between two adjacent patches. The learning rate is divided by 10 at epochs 8 and 11 and the total epoch is set to 12.

2) HRSC2016: HRSC2016 is the oriented ship detection dataset. There are 1061 samples in HRSC2016. 436 images are used for training and 181 images are used for validation. The testing set contains 444 samples.

Following the standard protocols [24], [30], we use the training and validation sets to train our models. We hold the original aspect ratios of each image and the shorter sides are resized to 800. The longer sides are no more than 1333. All our models are trained in 36 epochs and the learning rate is set to 0.005 and divided by 10 at 24 and 33 epochs.

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TABLE II
QUALITATIVE COMPARISONS AMONG DIFFERENT SINGLE SIMILARITY WEIGHT ON DOTA TESTING SET

Baseline	w_1	w_2	w_3	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
OPLD				89.86	82.24	53.02	67.49	74.43	76.96	86.46	90.77	83.28	85.63	58.45	64.84	73.23	67.18	58.75	74.18
	✓			90.28	84.75	54.19	69.11	74.94	74.43	86.28	90.84	83.17	86.42	56.40	64.19	70.99	67.25	59.67	74.19
		✓		89.56	82.21	54.49	70.03	75.62	75.86	87.25	90.85	84.22	86.20	61.23	64.29	73.40	69.10	51.90	74.42
			✓	89.35	82.57	55.67	71.53	77.38	75.85	87.22	90.86	83.36	86.11	56.91	63.83	70.14	69.25	53.69	74.25
ReDet				88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
	✓			88.86	84.32	55.41	78.63	77.56	83.72	87.96	90.83	87.45	85.10	61.61	63.85	76.81	71.44	60.00	76.90
		✓		89.07	85.29	54.55	75.78	77.40	81.76	87.76	90.88	85.81	87.08	66.63	69.12	75.17	70.88	59.40	77.04
			✓	89.63	84.91	53.76	76.46	77.47	81.81	87.87	90.87	86.07	87.13	66.56	69.22	75.15	69.54	62.14	77.24

TABLE III
QUALITATIVE COMPARISONS AMONG DIFFERENT COMBINED SIMILARITY WEIGHT ON DOTA TESTING SET

Baseline	w_1	w_2	w_3	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
OPLD	✓	✓		89.57	83.19	52.41	67.21	74.62	75.39	86.24	90.60	84.31	85.25	55.61	67.81	71.63	68.11	61.27	74.21
	✓		✓	89.35	82.57	55.67	71.53	77.38	75.85	87.22	90.86	83.36	86.11	56.91	63.83	70.14	69.25	53.09	74.21
		✓	✓	88.99	83.86	54.02	66.65	73.89	75.61	86.58	90.83	83.76	85.56	59.67	66.96	73.12	68.52	59.78	74.52
	✓	✓	✓	89.31	83.54	53.38	70.92	75.23	75.72	86.91	90.85	80.49	85.55	59.03	66.91	74.55	68.50	59.10	**74.67**
ReDet	✓	✓		88.05	83.50	52.70	73.01	72.29	84.48	88.10	90.84	88.16	86.73	66.59	63.28	77.56	69.03	72.35	77.11
	✓		✓	89.56	83.69	54.38	74.86	76.74	84.27	87.41	90.50	87.54	86.39	64.49	66.69	77.29	71.39	63.44	77.24
		✓	✓	89.63	84.91	53.76	76.46	77.47	81.81	87.87	90.87	86.07	87.13	66.56	69.22	75.15	69.54	61.54	77.20
	✓	✓	✓	88.85	84.20	60.34	75.45	75.65	84.37	88.21	90.91	87.97	85.63	59.78	64.91	78.31	80.32	65.20	**78.01**

3) NWPU VHR-10: NWPU VHR-10 [27] has 800 images of ten categories: airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. Annotations in [27] are labeled by horizontal bounding boxes. The (1/5), (1/5), and (3/5) of positive samples are used for training, validation, and testing. Considering that the number of images in NWPU VHR-10 and HRSC2016 are similar, we adopt the same settings to train our models on NWPU VHR-10.

B. Evaluation Criteria

For the DOTA dataset, standard mean Average Precision (mAP) under Pascal VOC 2007 criteria [43] is adopted as the evaluation metric to assess the performance of OOD. mAP under Pascal VOC 2007 and 2012 criteria are used to evaluate the performance of our SFFD on HRSC2016. Moreover, average precision (AP) over 0.5, 0.75, and 0.95 intersection over union (IoU) threshold (AP_{50}, AP_{75}, and AP_{95}) and AP under small-, medium-, and large-scale objects (AP_s, AP_m, and AP_l) are used to evaluate the effectiveness of our SFFD. For NWPU VHR-10, we further use AP, average recall (AR), and average F1-score (AF) to assess the performance of our method. Specifically, we also use the model parameters (M) and frames per second (FPS) (tasks per second) to evaluate the performance of our SFFD.

C. Ablation Studies

The ablation studies are constructed on the DOTA dataset. We merge the proposed CFF module into many OOD methods, including recent rotation-equivariant detector (ReDet) [24] and object-wise point-guided localization detector (OPLD) [58], to evaluate the performance of our SFFD. We also compare our method with one famous instance-level object relation-learning module (ORM) [39].

1) Settings: For OPLD [58], we adopt feature pyramid networks (FPNs) [59] integrated ResNet [60] as the backbone network. Five levels with strides of {4, 8, 16, 32, 64} with respect to the original image are introduced for FPN. Three anchor aspect ratios of {1:2, 1:1, 2:1} are used for RPN and the scale of anchors are set to $\{32^2, 64^2, 128^2, 256^2, 512^2\}$. We use SGD with a learning rate of 0.01. The weight decay and momentum are set to 0.0001 and 0.9. Four GPUs with a total of eight images per mini-batch are used for training. All our models on the DOTA dataset are trained for 12 epochs and the learning rate is decreased by 0.1 at epochs 9 and 11. For HRSC2016 and NWPU VHR-10 datasets, a total of 36 epochs are used for training and the learning rate is decreased by 0.1 at epochs 24 and 33.

For ReDet [24], a reimplemented ResNet [60] backbone, ReResNet, is used as the feature extractor. FPN with 15 anchors per location of each pyramid level is adopted in our experiments. We also use an SGD optimizer with momentum and weight decay of 0.9 and 0.0001 for training. Two GPUs with a total of four images per mini-batch are used for training and the learning rate is set to 0.005. For the DOTA dataset, we train all models for 12 epochs and the learning rate is decreased by 0.1 at epochs 9 and 11. For HRSC2016 and NWPU VHR-10 datasets, a total of 36 epochs are used for training and the learning rate is decreased by 0.1 at epochs 24 and 33. All the settings are reported in the original paper [24], [58].

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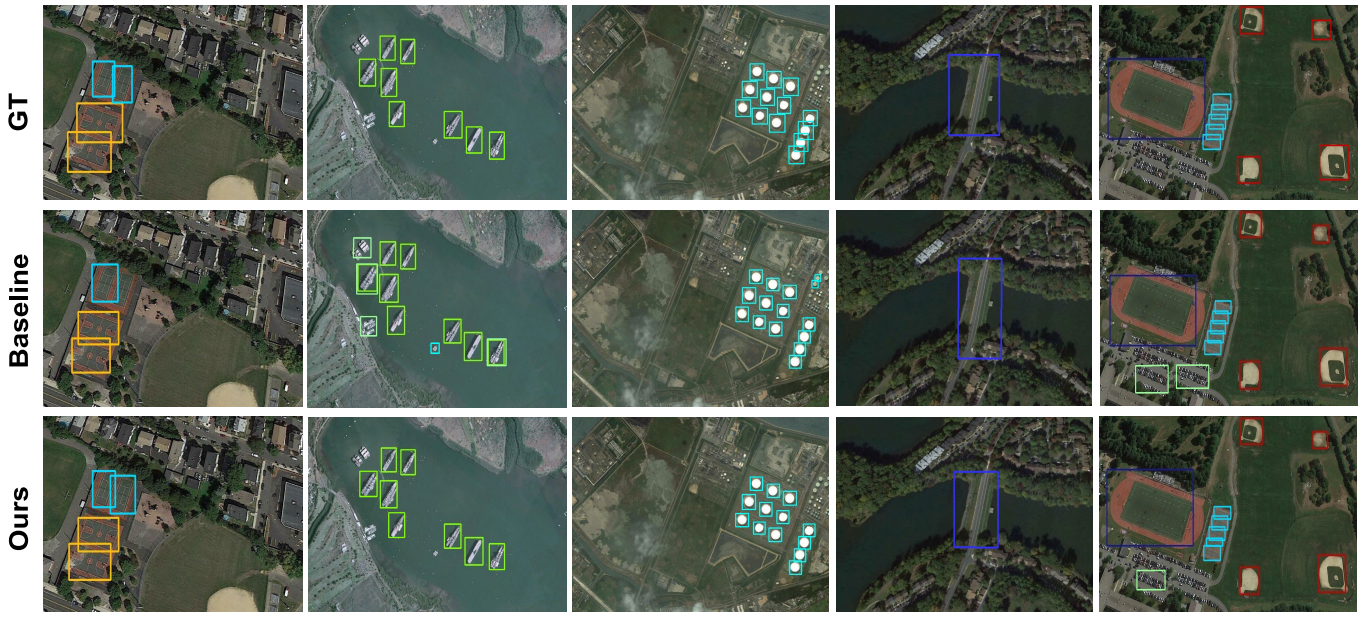


Fig. 6. Detection visualizations of baseline and our SFFD on the NWPU VHR-10 dataset. Compared to the ground truth, the baseline method miss-detects many targets and the proposed SFFD precisely detects more target instances.

TABLE IV
RESULTS OF DIFFERENT METHODS ON THE NWPU VHR-10 DATASET

Methods	SSD512 [42]	De-R-FCN [44]	DSOD300 [45]	RICADet [46]	RDAS512 [47]	MS-CNN [48]	ReDet [24]	FDOL [30]	SFFD (Ours)
mAP	78.40	79.10	79.80	87.12	89.50	89.60	90.14	90.68	**91.03**

TABLE V
AR, AF, AND AP COMPARISONS OF DIFFERENT METHODS ON THE NWPU VHR-AIRPLANE SUBDATASET

Methods	Trans CNN [27]	P-R [49]	RICNN [27]	YOLOv2 [41]	FourierHOG [50]	Double-P-R [49]	ORSIm [28]	FDOL [30]	SFFD (Ours)
AP	66.08	80.32	88.40	89.75	90.20	90.59	95.39	94.28 $\pm$ 4.00	**95.37** $\pm$ **3.86**
AR	–	–	–	84.92	87.78	–	91.12	90.05 $\pm$ 3.50	**91.28** $\pm$ **2.97**
AF	–	–	–	87.27	88.97	–	93.21	92.12 $\pm$ 3.74	**93.28** $\pm$ **3.38**

2) Gains of CFF: ORM [39] is a famous instance-level feature fusion method for object detection. Here, we compare our CFF with original OPLD [58], ReDet [24], and ORM-enhanced OPLD and ReDet methods. As illustrated in Fig. 5, compared with the base method, ORM enhanced method detects more missed small vehicles and ships. Importantly, our SFFD detects more targets and alleviates duplicate detection with different categories. As listed in Table I, the original OPLD with ResNet50 backbone extractor achieves 73.98 mAP. The ORM-improved OPLD framework boosts the detection mAP of base OPLD from 73.98 to 74.09 which demonstrates the effectiveness of excavating interaction among instance-level features for OOD. However, ORM only models the relationship among horizontal instances. Limited by the simplicity of CNN features, ORM is difficult to measure the orientation information for OOD. Thus, our CFF achieves 74.21 mAP which is 0.23 and 0.12 mAP higher than base OPLD- and ORM-enhanced methods. OPLD with ResNet101 backbone achieves 74.18 mAP. Our method still boosts OPLD with ResNet101 backbone by 0.49-mAP gains. Note that model parameters increased by ORM [39] and our proposed CFF are similar. However, the proposed CFF achieves competitive running times with different base methods (4.7 versus 5.7, 1.2 versus 2.1, and 4.9 versus 6.7) which brings out thoughts to our follow-up works.

TABLE VI
RESULTS OF DIFFERENT METHODS ON HRSC2016 DATASET

Methods	**Backbone**	**mAP (07)**	**mAP (12)**
RC2 [51]	–	75.70	–
RRPN [18]	R101	79.08	85.64
RoI Transformer [17]	R101	86.20	–
Gliding Vertex [52]	R101	88.20	–
PIoU [53]	DLA-34	89.20	–
DRN [54]	H-34	–	92.70
R^3Det [55]	R101	89.26	96.01
DAL [56]	R101	89.77	–
S^2A-Net [57]	R101	90.17	95.01
Oriented R-CNN [19]	R50	90.40	96.50
ReDet [24]	ReR50	90.46	97.63
Oriented R-CNN [19]	R101	90.50	97.60
SFFD (Ours)	ReR50	**90.52**	**97.68**

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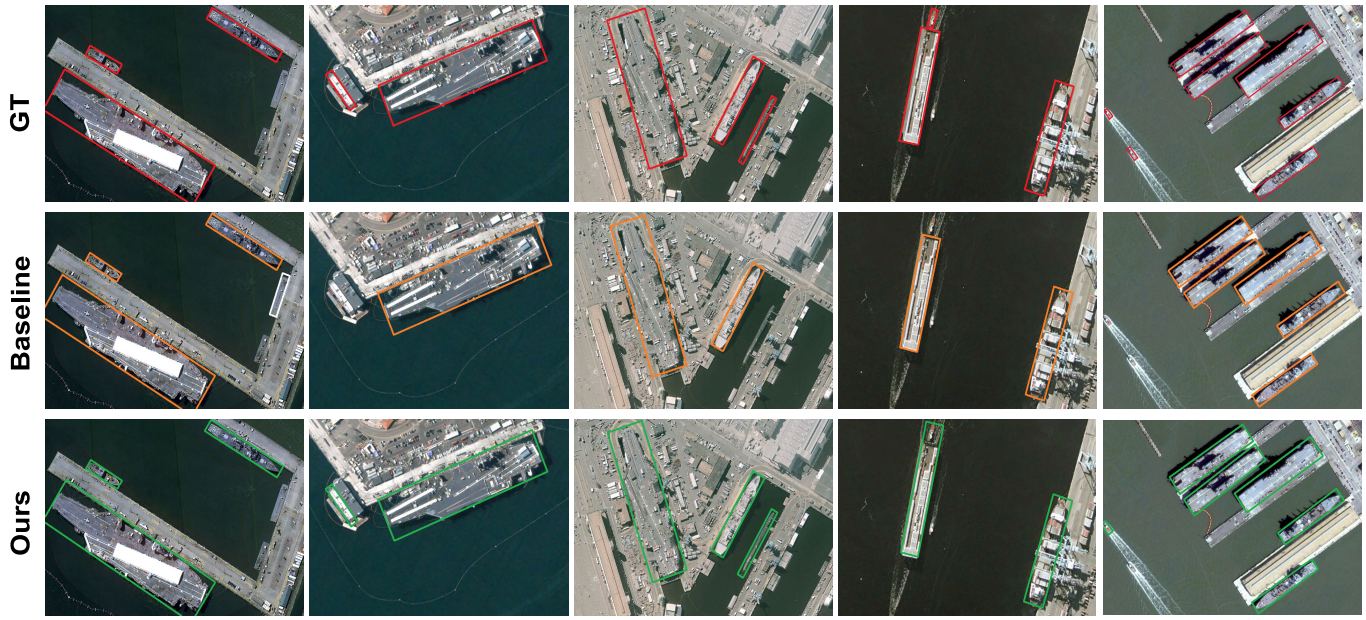


Fig. 7. Detection visualization of the baseline method and our SFFD on the HRSC2016 dataset. The red rectangle boxes denote the ground truth and the white box means the false detection. We visualize the detection results of the baseline method and our proposed SFFD in orange and green boxes. Compared to the ground truth, the baseline method miss-detects many targets and the proposed SFFD precisely covers more target ships.

TABLE VII
COMPARISONS OF DIFFERENT METHODS ON DOTA VALIDATION AND HRSC2016 TESTING SETS

Dataset		**DOTA (Training / Validation)**							**HRSC2016 (Trainval / Testing)**						
Methods	**Backbone**	**AP**$_{50}$	**AP**$_{75}$	**AP**$_{95}$	**AP**$_{05:95}$	**AP**$_s$	**AP**$_m$	**AP**$_l$	**AP**$_{50}$	**AP**$_{75}$	**AP**$_{95}$	**AP**$_{05:95}$	**AP**$_s$	**AP**$_m$	**AP**$_l$
OPLD [58]	R50	66.81 / 72.51*	43.38	1.98*	40.59	32.81	51.19	38.72	94.27 / 88.44*	86.51	1.29*	70.09	40.23	61.79	74.97
OPLD [58]	R101	65.24 / 74.68*	42.14	1.58*	40.19	32.97	52.13	38.02	94.02 / 88.23*	86.32	1.27*	69.86	40.19	61.08	74.68
FDOL [30]	ReR50	68.78 / 74.04*	42.01	1.32*	40.31	35.03	53.19	35.56	96.32 / 90.51*	88.64	1.44*	72.91	30.26	63.24	77.04
ReDet [24]	ReR50	69.21 / 75.24*	41.82	0.80*	40.83	35.39	53.78	37.46	96.30 / 90.46*	89.69	1.13*	73.87	50.11	65.23	77.56
SFFD (Ours)	ReR50	**69.54 / 75.68***	42.53	1.58*	**41.04**	**35.44**	53.39	37.72	**96.37 / 90.52***	89.68	**1.57***	**74.01**	**51.76**	64.37	77.53

The above-mentioned results demonstrate the considerable potential of modeling the interaction relation among multitype information on instance-level features.

We also unify ORM and the proposed CFF module into the ReDet framework which reimplements the ResNet50 backbone extractor with a rotation-equivariant network based on e2cnn [61]. The base ReDet network achieves 76.25 mAP with 6.7 testing FPS. The ORM method improves the performance of ReDet by a large margin, from 76.25 to 77.24 mAP. Moreover, our CFF still achieves the best mAP, which is 1.76 and 0.77 mAP higher than the base ReDet and ORM-improved ReDet networks. That is our CFF module significantly boosts different methods and achieves the best detection accuracy.

3) Effects of Similarity Weight: There are three types of similarity weights w_1, w_2, and w_3 in our SFFD. Here, we evaluate the contributions of different similarity weights to the detection task. Table II lists the detection results of single similarity weight enhanced methods. The w_1 integrated OPLD and ReDet achieve 0.01- and 0.65-mAP gains. This improvement demonstrates that homogeneously exploring the correlation among instance-level frequency features to enhance the representation of CNN RoI features is suboptimal for OOD in RSIs. w_2 and w_3 achieve the best performance on OPLD and ReDet, respectively. These results show that interactively exploring the internal relationship between frequency and CNN RoI features can significantly improve the performance of OOD.

Moreover, we report the detection results of different combined similarity weight-enhanced methods in Table III. $w_2 + w_3$ and $w_1 + w_3$ improve base OPLD and ReDet by 0.34 and 0.99 mAP, respectively, which achieve superior performance than single similarity weight-enhanced methods. Importantly, the comprehensive similarity weight $w_1 + w_2 + w_3$ significantly improves the base OPLD method from 74.18 to 74.67 mAP, which demonstrates that frequency features could assist CNN features to model the interaction relation among target instances in images. Similarly, experimental results on the ReDet network also prove our analysis. The best performance is achieved by integrating w_1, w_2, and w_3 into the base method and the $w_1 + w_2 + w_3$-integrated ReDet method achieves 78.01 mAP which is 1.76 mAP higher than the base ReDet.

D. Comparisons With State-of-the-Art Methods

We compare our SFFD method with lots of object detection methods on NWPU VHR-10, HRSC2016, and DOTA datasets. Most of them are published in the last three years.

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TABLE VIII
QUANTITATIVE RESULTS OF DIFFERENT OBJECT DETECTION MODELS ON DOTA. RESULTS OF METHODS ARE PARTLY OBTAINED FROM THEIR RELEASED CODE AND RECONSTRUCTED WHEN NEEDED

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
single-scale:																	
FR-O [62]	R101	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	54.13
ICN [63]	R101	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
CADNet [20]	R101	87.80	82.40	49.40	73.50	71.10	63.50	76.60	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
RoI Transformer [17]	R101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
DRN [54]	H-104	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
CenterMap [64]	R50	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.7	71.74
SCRDet [65]	R101	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
FAOD [66]	R101	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
R^3Det [55]	R152	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
S^2A-Net [57]	R50	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
OPLD [58]	R101	89.86	82.24	53.02	67.49	74.43	76.96	86.46	90.77	83.28	85.63	58.45	64.84	73.23	67.18	58.75	74.18
Oriented R-CNN [19]	R50	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
ReDet [24]	ReR50	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
Oriented R-CNN [19]	R101	88.86	83.48	55.27	76.92	74.27	82.10	87.52	90.90	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
ORM [39]	ReR50	89.32	83.59	55.35	74.93	76.15	83.85	87.51	90.69	86.22	85.77	63.83	65.40	77.19	71.23	59.99	76.73
FDOL [30]	ReR50	88.82	84.15	56.25	75.04	74.89	83.88	88.14	90.91	87.97	85.42	59.44	66.36	77.05	74.11	65.89	77.22
SFFD (Ours)	ReR50	87.91	83.19	53.87	78.37	73.90	85.30	88.58	90.81	87.67	87.07	66.03	62.53	77.89	76.46	70.60	**78.01**
multi-scale:																	
FR-O [62]	R101	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05
DRN [54]	H-104	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	73.23
FAOD [66]	R101	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
Gliding Vertex [52]	R101	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
CenterMap [64]	R101	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
OWSR [67]	R101	90.41	85.21	55.00	78.27	76.19	72.19	82.14	90.70	87.22	86.87	66.62	68.43	75.43	72.70	57.99	76.36
OPLD [58]	R101	89.37	85.82	54.10	79.58	75.00	75.13	86.92	90.88	86.42	86.62	62.46	68.41	73.98	68.11	63.69	76.43
S^2A-Net [57]	R50	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58	79.42
ReDet [24]	ReR50	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	74.67	80.10
GWD [68]	R152	89.66	84.99	59.26	82.19	78.97	84.83	87.70	90.21	86.54	86.85	73.47	67.77	76.92	79.22	74.92	80.23
SFFD (Ours)	ReR50	89.12	82.50	61.99	81.19	77.85	86.20	88.20	90.87	88.02	84.38	68.42	68.02	84.44	81.00	72.51	**80.31**

1) Experiments on NWPU VHR-10 Dataset: The comparisons with many other methods on NWPU VHR-10 are listed in Table IV. ReDet [24] and FDOL [30] achieve 90.14 and 90.68 mAP, respectively. Our SFFD achieves the best detection result which is 0.89 and 0.35 mAP higher than the best two competitors. Fig. 6 illustrates detection visualization of the baseline method and our SFFD. We can clearly see that the base method miss-detects many targets. The proposed SFFD covers more precise target instances than the baseline approach.

In order to compare our method with other spatial-frequency features fusion approaches [28], [30], we focus on detecting objects with high rotations on NWPU VHR-10. Followed by method [27], NWPU VHR-10 is randomly divided into training, validation, and testing sets. Here, NWPU VHR-10 is partitioned several times to evaluate the stability of our method. As listed in Table V, ORSIm achieves 95.39, 91.12, and 93.21 AP, AR, and AF, respectively, which also proves the effectiveness of exploring interaction relation between frequency and CNN features. FDOL [30] constructs an orientation weight from the frequency domain to boost the orientation sensitivity of CNN features which achieves 94.28, 90.05, and 92.12 AP, AR, and AF, respectively. However, FDOL unilaterally evaluates the influence of frequency information on spatial features which ignores the interaction relationship between the two sides. Thus, our method achieves the best AP, AR, and AF of 95.37, 91.28, and 93.28, respectively.

2) Experiments on the HRSC2016 Dataset: Here, we report the comparisons between our SFFD and other OOD methods on the HRSC2016 dataset. As listed in Table VI, the best competitor Oriented R-CNN [19] with ResNet101 backbone achieves 90.50 and 97.60 mAP under Pascal VOC 2007 and 2012 evaluation criteria. Our proposed SFFD achieves the best performance with mAP of 90.52 and 97.68 under Pascal VOC 2007 and 2012 evaluation criteria which demonstrates the effectiveness of modeling the interaction relation between frequency and CNN features. Moreover, we further evaluate the detection performance of different methods under COCO [69] criteria in Table VII. Note that $*$ denotes that the detection results are calculated under Pascal VOC criteria. Considering that the proposed L-FDA module extracts multiscale frequency features which could improve the performance of small object detection. Thus, our SFFD achieves the superior AP_s of 51.76, which is 1.65 higher than the best competitor ReDet [24].

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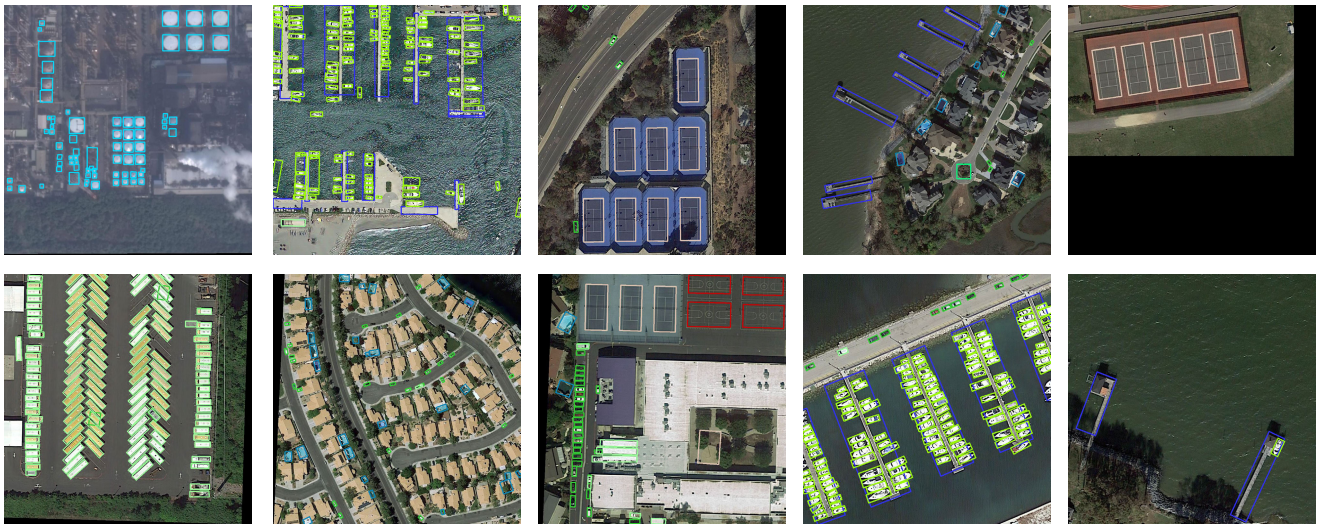


Fig. 8. Detection visualizations of our SFFD on the DOTA dataset.

Our method also achieves the best AP_{50}, AP_{95}, and $AP_{.5:95}$ of 96.37, 1.57, and 74.01, respectively. Fig. 7 illustrates the detection results of the baseline method and our SFFD on the HRSC2016 dataset. The ground truth of bounding boxes is visualized in red rectangle boxes. The detection results of the baseline method and our SFFD are visualized in orange and green rectangle boxes. We can clearly see that compared with the baseline method, the proposed SFFD detects more missdetected ships.

3) Experiments on the DOTA Dataset: The comparisons with many SOTA methods on the DOTA dataset are listed in Tables VII and VIII. We first compare our SFFD with many OOD methods by training all models on the DOTA training set and evaluating the performance on the validation set. Compared with OPLD [58], FDOL [30] and our SFFD utilize multiscale frequency features to enhance the orientation sensitivity of CNNs which achieves 35.03 and 35.44 AP_s. The best competitor ReDet [24] achieves 69.21, 41.82, and 0.80 AP_{50}, AP_{75}, and AP_{95}, respectively. Importantly, our SFFD achieves the best performance among most evaluation criteria which demonstrates the effectiveness of fusing multitype features for OOD. For the single-scale training and testing method in Table VIII, Oriented R-CNN [19] with ResNet101 backbone extractor tries to design high-quality rotated proposals to better represent oriented instances and achieves 76.28 mAP. The ReDet [24] constructs a novel rotation-equivariant ResNet backbone network to model the rotation-invariance features and achieves 76.25 mAP. The best competitor FDOL [30] achieves 77.22 mAP which is superior to all other methods. Moreover, our SFFD explores the interaction relation among instance-level spatial-frequency features and achieves the best 78.01 mAP which is 0.79% higher than the best competitor FDOL. After adopting multiscale training and testing strategy, the performance of all the methods has improved by a large margin. The ReDet method [24] improves from 76.25 to 80.10 mAP, which achieves 3.85 gains. Our SFFD achieves the best performance, which is 0.21 and 0.08 mAP higher than ReDet and the best competitor Gaussian Wasserstein distance (GWD) [68] methods. Fig. 8 illustrates the detection results of our SFFD on the DOTA dataset. The proposed SFFD accurately detects different scales of oriented objects in RSIs.

V. Conclusion

This article proposes an instance-aware SFFD for OOD in RSIs. First, an L-FDA module is built along with CNNs layers to extract frequency features. Compared with CNNs, the proposed L-FDA locates the mutational signals from the frequency space. The orientation information recorded in these signals can be accurately reconstructed to the original images by the inverse transform operation. The CNN and frequency features are then sent into the RoI Pooling layer to achieve the instance-level RoI features. Subsequently, a novel instance-aware CFF module is introduced to model the interaction relation among these instance-level multitype features. Three types of similarity weights are constructed for providing an explicit indicator to compensate for the orientation information ignored by instance-level CNN features. Finally, we unify the proposed L-FDA and CFF modules into a detector (SFFD) to detect the oriented objects in RSIs. Experimental results demonstrate that the proposed SFFD achieves SOTA detection accuracy. Furthermore, we will explore more forms of spatial-frequency feature fusion within the detection framework to improve the performance of OOD.

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